1. Overview

This data wrangling process aims to prepare an analysis-ready encounter-level dataset for answering the research questions. The final dataset will contain a binary outcome variable for postoperative hypoxemia, patient demographics (age, sex, BMI), and the latest available preoperative results for ten specific laboratory tests, namely, leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, and potassium.

2. Source Data Tables and Relevant Features

Relevant information on patient demographics, lab test results and postoperative hypoxemia status are obtained from three primary data tables from the MOVER dataset [1]. The relevant variables from these tables are outlined in the Data Mapping section. All three data tables are at the encounter level, and share the same unique identifiers for each encounter for each patient (namely, the log id (LOG_ID) and medical record number (MRN)).

2.1 Patient Postoperative Complications Table

The postoperative complications table contains information on 56,127 encounters. The primary variable of interest within this table is SMRTDTA_ELEM_VALUE, which contains the specific names of postoperative complications.

2.2 Patient Information Table

The patient information table contains information on 65,728 encounters. The descriptions of variables of interest are listed below:

1. BIRTH_DATE: the patient's age in years.
2. SEX: the patient's genotypic sex.
3. HEIGHT: the patient's height in feet and inches (ex. 5'6).
4. WEIGHT: the patient's weight in ounces.
5. AN_START_DATETIME: the start date and time of anesthesia.

We will rename the BIRTH_DATE variable to AGE for clarity.

2.3 Patient Labs Table

The patient labs table contains information for 29,079,344 lab test results across 56,127 unique encounters. The table contains duplicated rows, which will be dropped (leaving only one row per unique instance) during the analysis. The variables of interest are

1. Lab Code: the unique identifier for the laboratory test.

2. Lab Name: descriptive common name for the laboratory test.
3. Observation Value: observed value (results) for the lab.
4. Measurement Units: units associated with the observed value.
5. Collection Datetime: date and time at which the lab was taken.

3. Data Integration

After selecting the relevant features, we will create three separate dataframes from each of the source tables:

1. Postoperative hypoxemia table: containing log id, MRN, and postoperative complications.
2. Demographics table: with the columns log id, MRN, age, sex, and anesthesia start time.
3. Lab test results table: with the columns log id, MRN, lab code, lab name, value and unit of the results, and collection time.

These three tables will then be merged into a comprehensive dataframe using log id and MRN.

4. Data Cleaning and Feature Engineering

4.1 Outcome Variable: Binary Indicator of Postoperative Hypoxemia

We will use the SMRTDTA_ELEM_VALUE (name of postoperative complications) variable to create the binary indicator for postoperative hypoxemia. Entries containing the keyword "hypoxemia" are classified as positive cases, while all other entries are considered negative.

The SMRTDTA_ELEM_VALUE variable has 6,741 out of 203,945 entries being non-null. We assume that these missing entries correspond to cases where no specific postoperative complications were observed. Thus, missing entries are also treated as negative cases.

To ensure the correct identification of positive cases, all text entries in this field will first be converted to lowercase. We will then create a binary variable that assigns a value of 1 to rows where the lowercase SMRTDTA_ELEM_VALUE contains the term "hypoxemia" and a value of 0 otherwise.

4.2 Demographic Features

We will calculate the patient's BMI as weight in kilograms divided by height in meters squared [2]. To facilitate this calculation, we will first treat the missing values in height and weight, then convert height from feet and inches into meters and weight from ounces into kilograms.

Missing values for height and weight are addressed using a two-step procedure. Because a single patient can have multiple encounters, we first attempt to fill missing heights and weights using records from the patient's nearest available encounter. If no available data exists, we plan to use regression imputation to predict the missing values based on the patient's age, sex, and whichever height or weight measurement is available.

Before attempting imputation, we will visually inspect the missingness pattern. If the data appears to be MNAR, we consider retaining only the complete cases for subsequent analysis.

There are no missing values in the age and sex fields.

4.3 Timestamps for Anesthesia and Lab Tests

The anesthesia start time and lab test results collection time are both stored as object variables, which will need to be converted into datetime format to allow for accurate filtering and comparison during subsequent wrangling. For the same purpose, we will only include encounters and lab test instances with non-missing anesthesia star time and collection time, respectively.

4.4 Lab Test Results

We will follow the outlined procedure to select relevant laboratory tests. For each encounter, the merged data is filtered to include only those tests conducted prior to the anesthesia start time. We then filter the lab results to include only the ten lab tests of interest based on the Lab Name variable. For lab types with multiple names, we select only the most common lab names. For example, the Leukocytes test has entries such as "Leukocytes" or "Leukocytes^^corrected for nucleated erythrocytes" under the Lab Name variable; we filter based on the most common version, which is the latter. Finally, the latest available instance of each lab test result prior to surgery is selected and stored in the final dataset.

For the observed test results and measurement units, values recorded as 999999 and "unknown," respectively, are considered missing. We will skip these specific lab test instances to ensure that only records with complete information are retained.

5. Citations

(To be finalized in the final version)

1. MOVER data documentation: https://mover.ics.uci.edu/documentation.html
2. BMI reference: https://my.clevelandclinic.org/health/articles/9464-body-mass-index-bmi